

Overview:

I need a small API integration service built in Node.js that pulls weather data from a free public API, processes it, and exposes a simplified endpoint our other tools can consume. Basically a middleware layer that reformats and caches weather data.

Scope of work:

- Node.js + Express service
- On startup (and every 30 minutes via a scheduled job), fetch current weather data from OpenWeatherMap's free API for 5 hardcoded city names
- Store fetched data in an in-memory cache (a simple JS object or lightweight library like node-cache)
- Expose the following custom endpoints:
 - ``GET /weather`` — returns all 5 cities with: city name, temperature in both Celsius and Fahrenheit, humidity, wind speed, a simple condition label (Sunny/Cloudy/Rainy/Snowy based on the weather code), and the timestamp of last fetch
 - ``GET /weather/:city`` — returns data for 1 city
 - ``GET /weather/compare?cities=cityA,cityB`` — returns a side-by-side comparison object for 2 cities
 - ``GET /health`` — returns uptime and last successful fetch time

References:

Resources:

- OpenWeatherMap freemium plan API:
<https://openweathermap.org/appid>
- OpenWeatherMap API docs: <https://openweathermap.org/current>

Additional information:

Additional Requirements:

- If external API is down, serve stale cached data and include a `"stale": true` flag in the response
- Rate limiting: max 30 requests per minute per IP using `express-rate-limit`
- Include a `.env.example` file showing where to put the API key
- Cities to use: Oslo, Lisbon, Nairobi, Osaka, Medellín
- Response format should be clean and flat — no deeply nested objects
- Deploy the live prototype to a free hosting service like Vercel, Netlify, or GitHub Pages
- NO AI SLOP. Send daily work-in-progress screenshots

Required Deliverables:

1. Link to the live hosted site.
 2. Zip file containing the clean, commented source code.
 3. README document explaining where and how to swap out the free API key
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- I've never set up an API key before so please make the README really clear. Also, the compare endpoint is important to us. Make sure it calculates the difference in temperature and humidity between the 2 cities and includes that in the response.